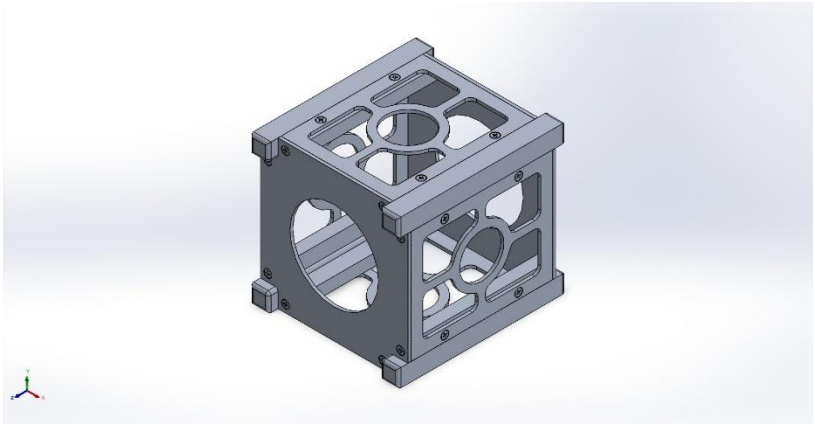
Politecnico di Torino



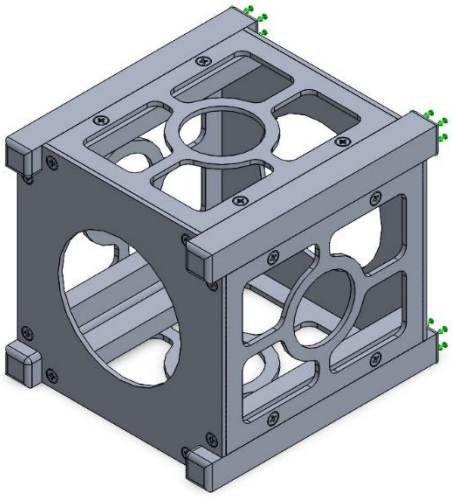
Simulazione di CubeSat

Data: martedì 27 ottobre 2026
Designer: Cristian Caruso
Nome studio: Frequenza 1
Tipo di analisi: Frequenza

Sommario

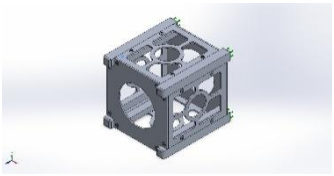
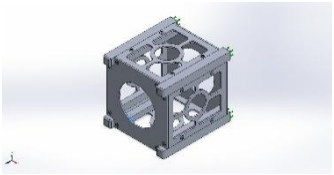
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Proprietà studio 10
Unità 11
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Carichi e fissaggi 15
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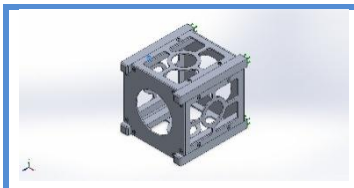
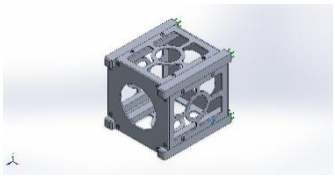
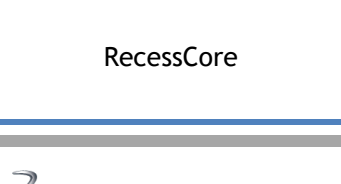
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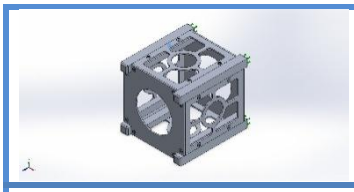
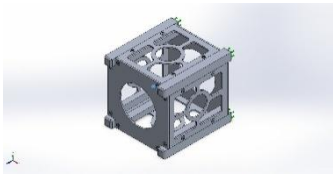


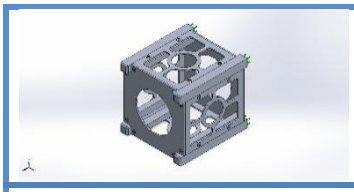
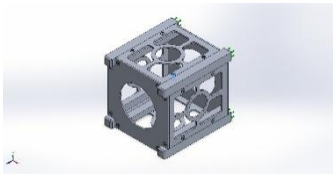
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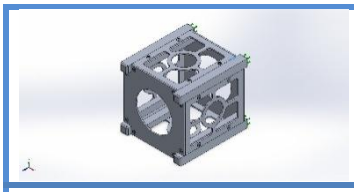
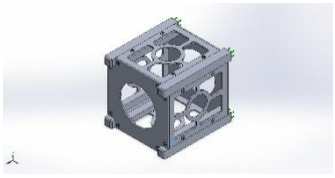
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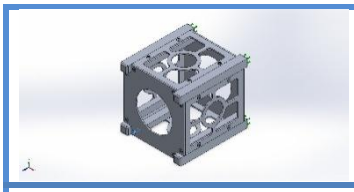
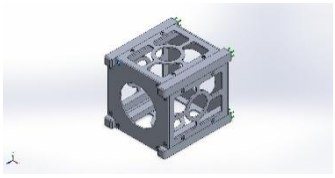
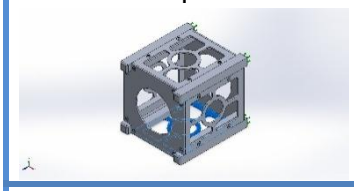
Nome documento e Riferimento	Trattato come	Proprietà volumetriche	Percorso/Data documento modificati
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RecessCore	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3**	**C:\SOLIDWORKS Data\browser\ISO\bolts**

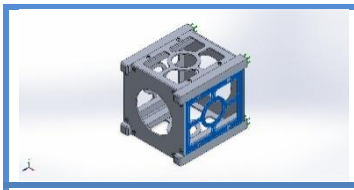
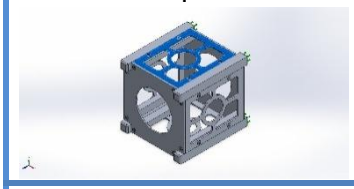
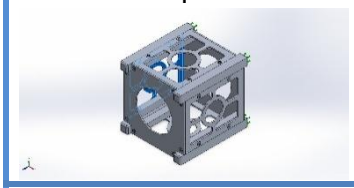
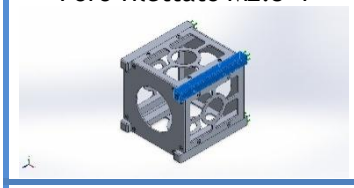
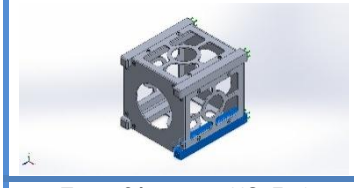
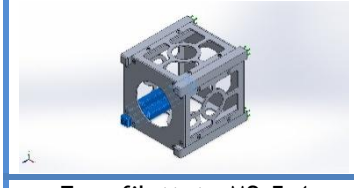
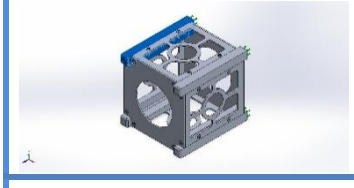
		**Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
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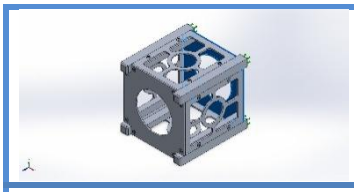
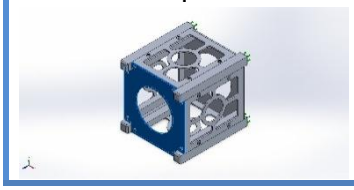
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Foro svasato per Vite M2.5-1 	Corpo solido	**Massa:0,0181299 kg Volume:6,71478e-06 m^3 Densità:2.700 kg/m^3 Spessore:0,177673 N**	**C:\Users\carus\Desktop\CubeSat\lateral_panel.SLDPRT** Oct 27 00:37:28 2026
Foro svasato per Vite M2.5-1	Corpo solido	**Massa:0,0181299 kg Volume:6,71478e-06 m^3 Densità:2.700 kg/m^3 Spessore:0,177673 N**	**C:\Users\carus\Desktop\CubeSat\lateral_panel.SLDPRT** Oct 27 00:37:28 2026

			
Foro svasato per Vite M2.5-1 	Corpo solido	**Massa:0,0181299 kg Volume:6,71478e-06 m^3 Densità:2.700 kg/m^3 Spessore:0,177673 N**	**C:\Users\carus\Desktop\CubeSat\lateral_panel.SLDPRT** Oct 27 00:37:28 2026
Foro svasato per Vite M2.5-1 	Corpo solido	**Massa:0,0181299 kg Volume:6,71478e-06 m^3 Densità:2.700 kg/m^3 Spessore:0,177673 N**	**C:\Users\carus\Desktop\CubeSat\lateral_panel.SLDPRT** Oct 27 00:37:28 2026
Foro filettato M2.5-4 	Corpo solido	**Massa:0,0846295 kg Volume:3,13443e-05 m^3 Densità:2.700 kg/m^3 Spessore:0,829369 N**	**C:\Users\carus\Desktop\CubeSat\rail.SLDPRT** Oct 27 01:08:46 2026
Foro filettato M2.5-4 	Corpo solido	**Massa:0,0846295 kg Volume:3,13443e-05 m^3 Densità:2.700 kg/m^3 Spessore:0,829369 N**	**C:\Users\carus\Desktop\CubeSat\rail.SLDPRT** Oct 27 01:08:46 2026
Foro filettato M2.5-4 	Corpo solido	**Massa:0,0846295 kg Volume:3,13443e-05 m^3 Densità:2.700 kg/m^3 Spessore:0,829369 N**	**C:\Users\carus\Desktop\CubeSat\rail.SLDPRT** Oct 27 01:08:46 2026
Foro filettato M2.5-4 	Corpo solido	**Massa:0,0846295 kg Volume:3,13443e-05 m^3 Densità:2.700 kg/m^3 Spessore:0,829369 N**	**C:\Users\carus\Desktop\CubeSat\rail.SLDPRT** Oct 27 01:08:46 2026
Foro svasato per Vite M2.5-1	Corpo solido	**Massa:0,0351387 kg Volume:1,30143e-05 m^3 Densità:2.700 kg/m^3**	**C:\Users\carus\Desktop\CubeSat\top_down_plate.SLDPRT**

		**Spessore:0,344359 N**	Oct 27 00:12:19 2026
Foro svasato per Vite M2.5-1 	Corpo solido	**Massa:0,0351387 kg** **Volume:1,30143e-05 m^3** **Densità:2.700 kg/m^3** **Spessore:0,344359 N**	**C:\Users\carus\Desktop\CubeSat\top_down_plate.SLDPRT** Oct 27 00:12:19 2026

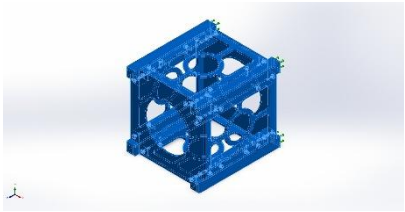
Proprietà studio

Nome studio	Frequenza 1
Tipo di analisi	Frequenza
Tipo di mesh	Mesh di elementi solidi
Numero di frequenze	5
Annulla accoppiamento modalità corpo libero miste	Disattivo
Tipo di solver	FFEPlus
Molla morbida:	Disattivo
Opzioni di unione non congruenti	Automatica
Opzione termica	Includi carichi di temperatura
Temperatura di deformazione zero	298 Kelvin
Includi effetti della pressione fluida da SOLIDWORKS Flow Simulation	Disattivo
Cartella Risultato	Documento SOLIDWORKS (C:\Users\carus\Desktop\CubeSat)

Unità

Sistema di unità:	SI (MKS)
Lunghezza/Spostamento	mm
Temperatura	Kelvin
Velocità angolare	Rad/sec
Pressione/Sollecitazione	N/m^2

Proprietà materiale

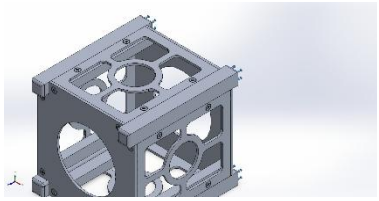
Riferimento modello	Proprietà		Componenti
	**Nome:**	6061-T6 (SS)	CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-1), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-10), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-11), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-12), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-13), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-14), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-15), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-16), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-17), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-18), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-19),
	Tipo di modello:	Isotropico elastico lineare	
	Fallimento di default:	Sollecitazione massima von Mises	
	Snervamento:	2,75e+08 N/m^2	
	Resistenza alla trazione:	3,1e+08 N/m^2	
	Densità di massa:	2.700 kg/m^3	
	Modulo elastico:	6,9e+10 N/m^2	
	Coefficiente di Poisson:	0,33	
	Coefficiente di espansione termica:	2,4e-05 /Kelvin	

		CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-2), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-20), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-21), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-22), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-23), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-24), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-25), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-26), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-27), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-28), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-29), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-3), CorpoSolido 1(RecessCore)(countersunk

		flat head cross recess screw_iso-30), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-31), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-32), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-4), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-5), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-6), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-7), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-8), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-9), CorpoSolido 1(Foro svasato per Vite M2.5-1)(lateral_panel-1), CorpoSolido 1(Foro svasato per Vite M2.5-1)(lateral_panel-2), CorpoSolido 1(Foro svasato per Vite M2.5-1)(lateral_panel-3), CorpoSolido 1(Foro svasato per Vite M2.5-1)(lateral_panel-4), CorpoSolido 1(Foro filettato M2.5-4)(rail-1), CorpoSolido 1(Foro filettato M2.5-4)(rail-2),

		CorpoSolido 1(Foro filettato M2.5-4)(rail-3), CorpoSolido 1(Foro filettato M2.5-4)(rail-4), CorpoSolido 1(Foro svasato per Vite M2.5-1)(top_down_plate-1), CorpoSolido 1(Foro svasato per Vite M2.5-1)(top_down_plate-2)
Dati curva:N/A		

Carichi e fissaggi

Nome fissaggio	Immagine fissaggio	Dettagli fissaggio
Fisso-1		**Entità:** 4 facce **Tipo:** Geometria fissa

Informazioni relative al mesh

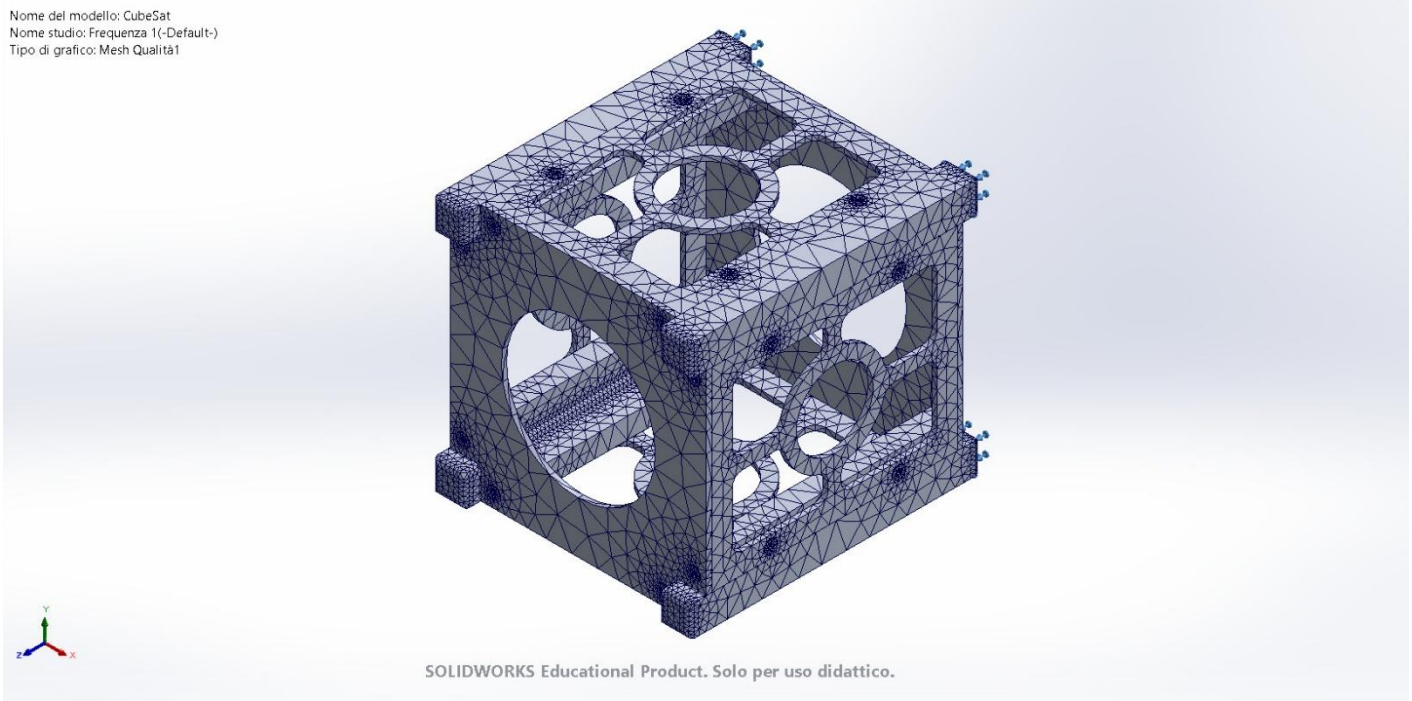
Tipo di mesh	Mesh di elementi solidi
Mesher usato:	Mesh basata su curvatura di blend
Punti jacobiani per mesh di alta qualità	16 Punti
Dimensione massima elemento	13,0766 mm
Dimensione elemento minima	0,653832 mm
Qualità della mesh	Ottima
Ripeti mesh parti non riuscite in modo indipendente	Disattivo
Riutilizza mesh per parti uguali in un assieme (solo mesher basato su curvatura di blend)	Disattivo

Informazioni relative al mesh - Dettagli

Nodi totali	197469
Elementi totali	115468
Rapporto d'aspetto max	15,072
% di elementi con rapporto d'aspetto < 3	91,9
Percentuale di elementi con rapporto d'aspetto > 10	0,0909
Percentuale di elementi distorti	0
Tempo per completare la mesh(hh;mm;ss):	00:00:10
Nome del computer:	

Grafici qualità della mesh

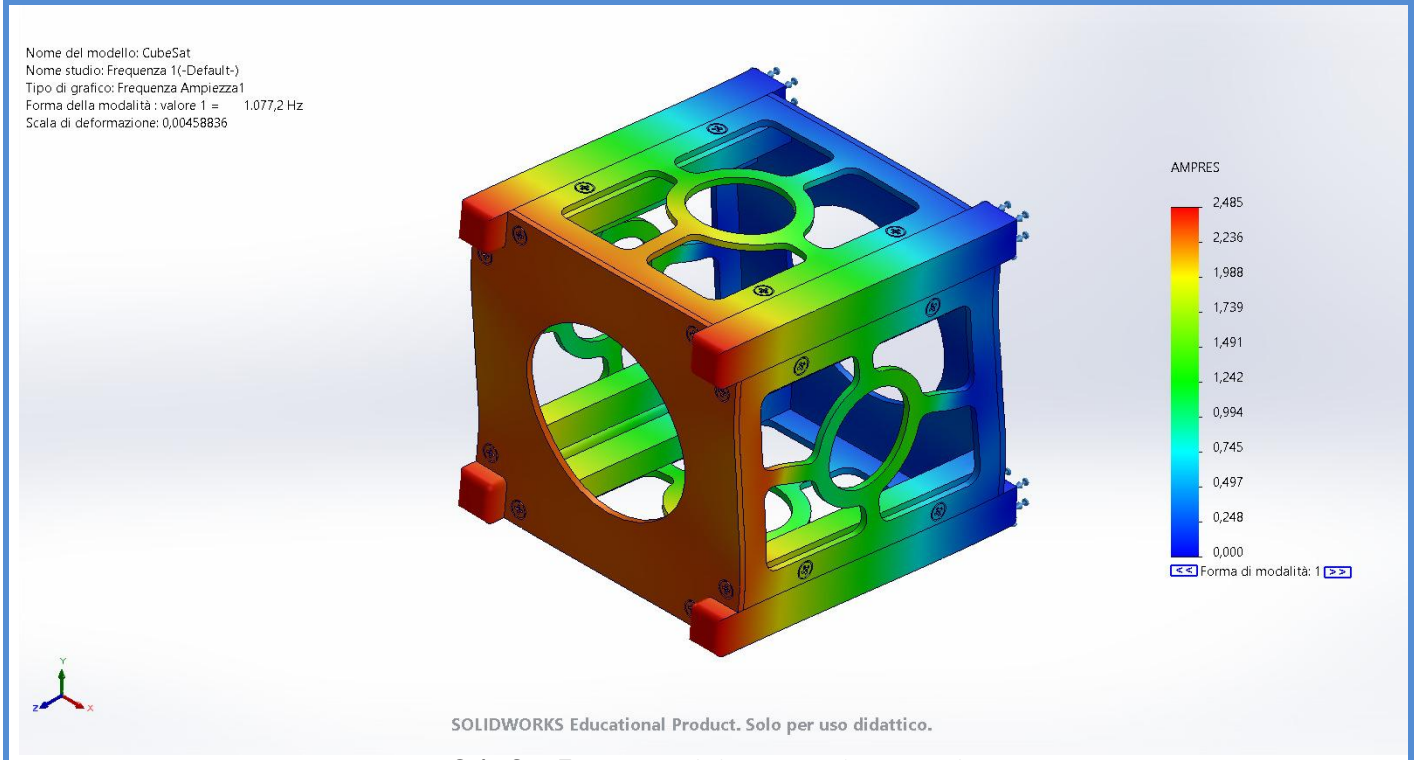
Nome	Tipo	Min	Max
Qualità1	Mesh	-	-



CubeSat-Frequenza 1-Qualità-Qualità1

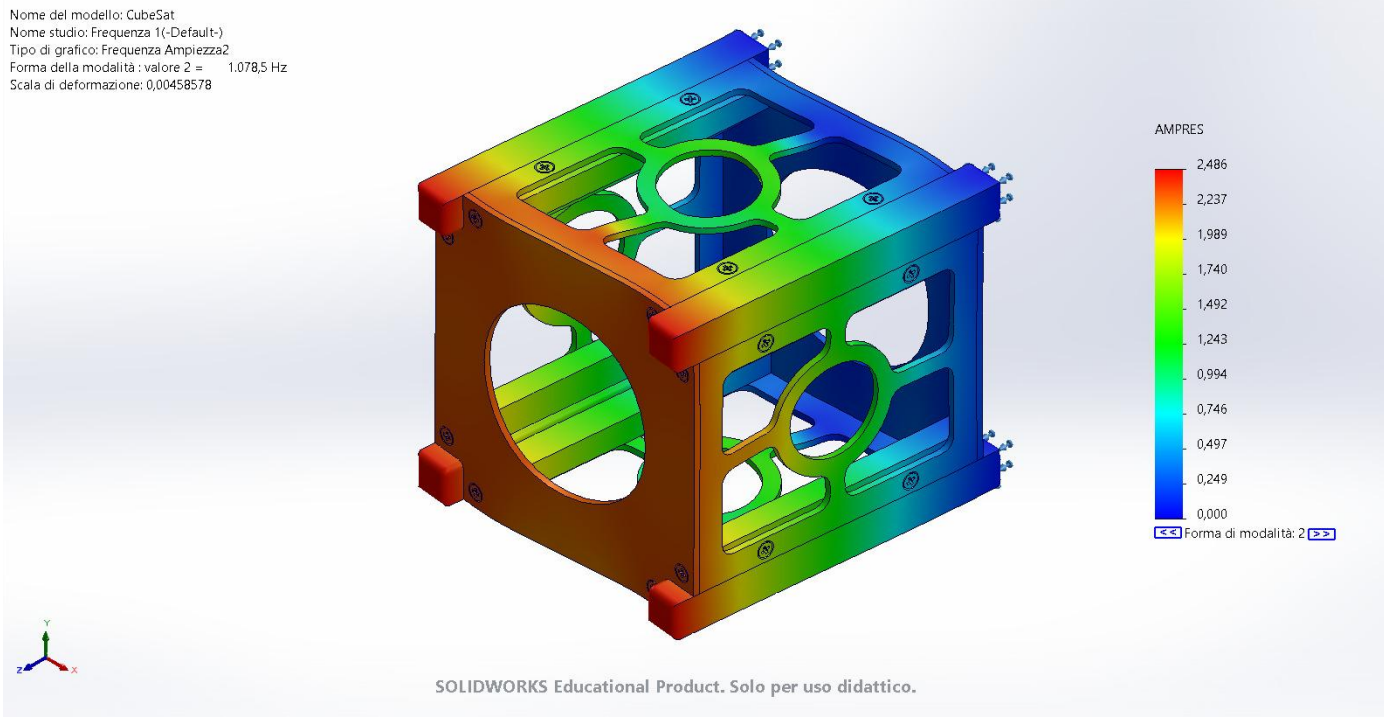
Risultati studio

Nome	Tipo	Min	Max
Ampiezza1	AMPRES: ampiezza risultante Grafico per Sagoma modale: 1(Valore = 1.077,21 Hz)	0,000 Nodo: 67439	2,485 Nodo: 96607



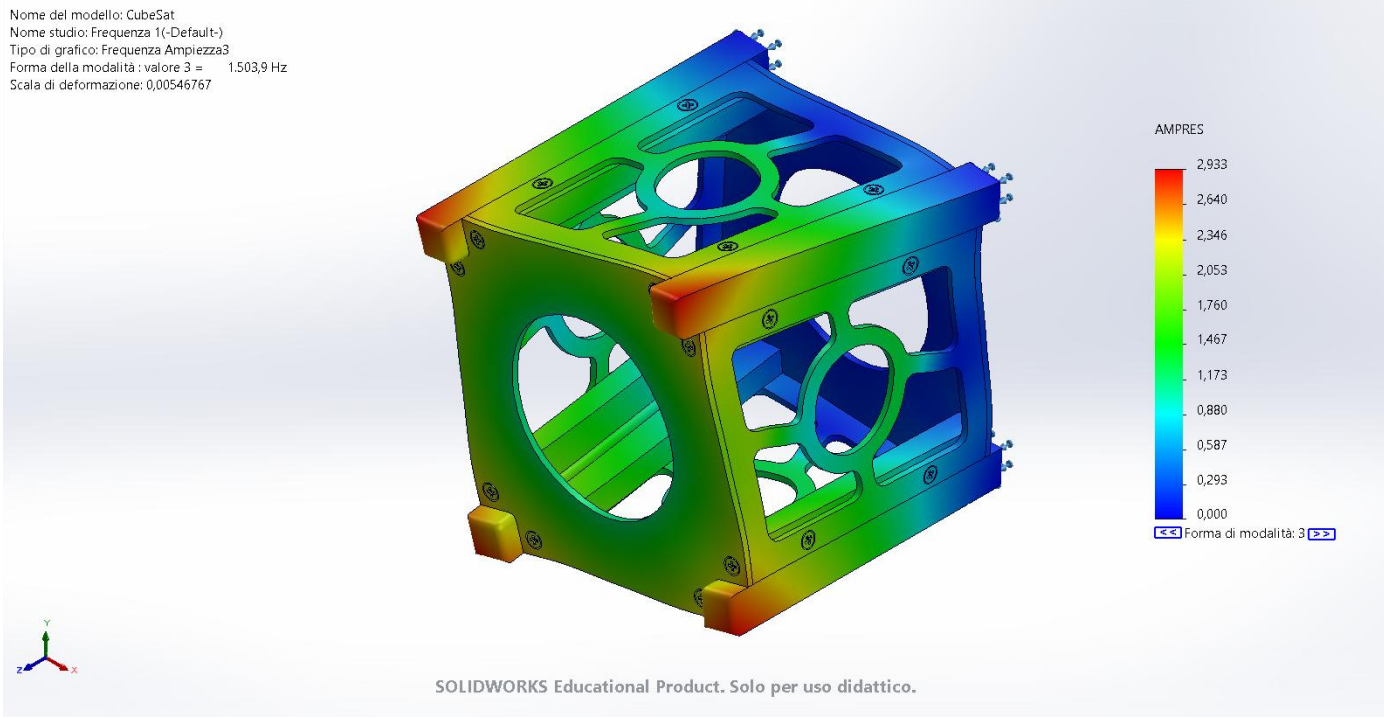
CubeSat-Frequenza 1-Ampiezza-Ampiezza1

Nome	Tipo	Min	Max
Ampiezza2	AMPRES: ampiezza risultante Grafico per Sagoma modale: 2(Valore = 1.078,47 Hz)	0,000 Nodo: 67439	2,486 Nodo: 67437



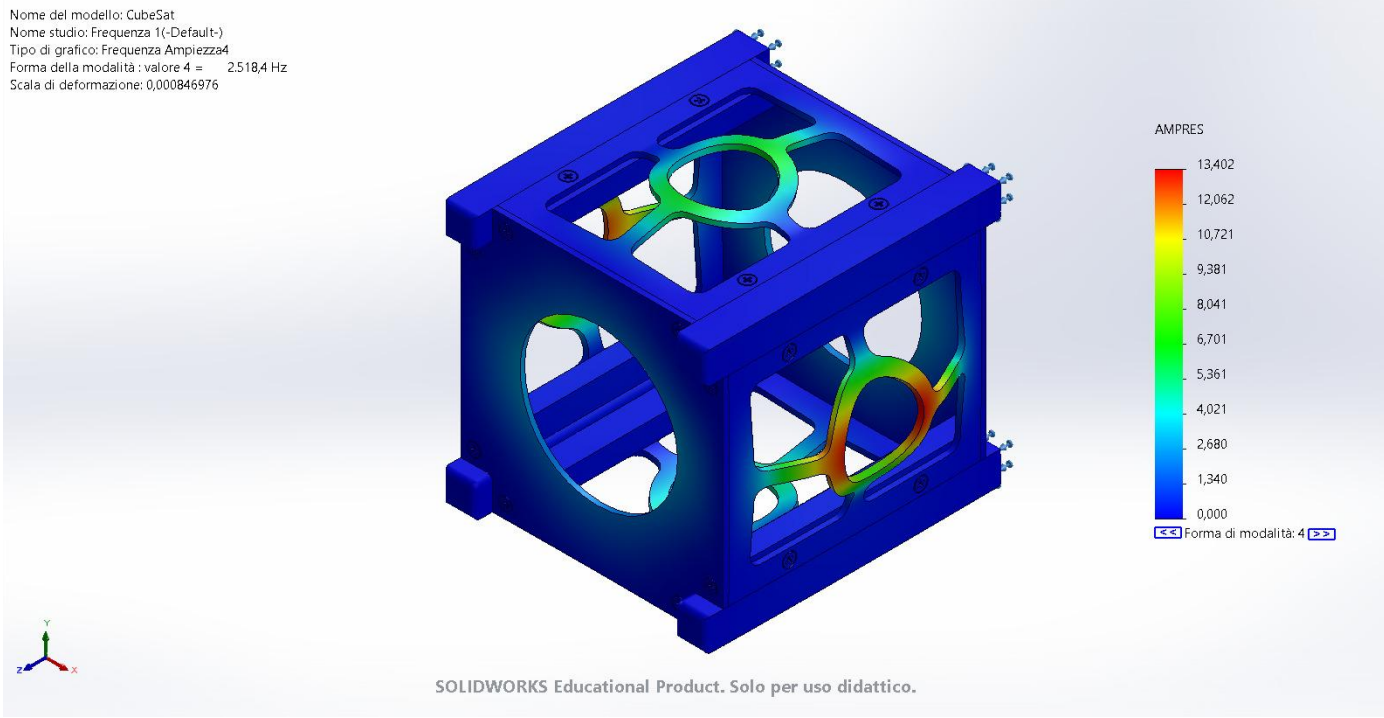
CubeSat-Frequenza 1-Ampiezza-Ampiezza2

Nome	Tipo	Min	Max
Ampiezza3	AMPRES: ampiezza risultante Grafico per Sagoma modale: 3(Valore = 1.503,94 Hz)	0,000 Nodo: 67439	2,933 Nodo: 126714



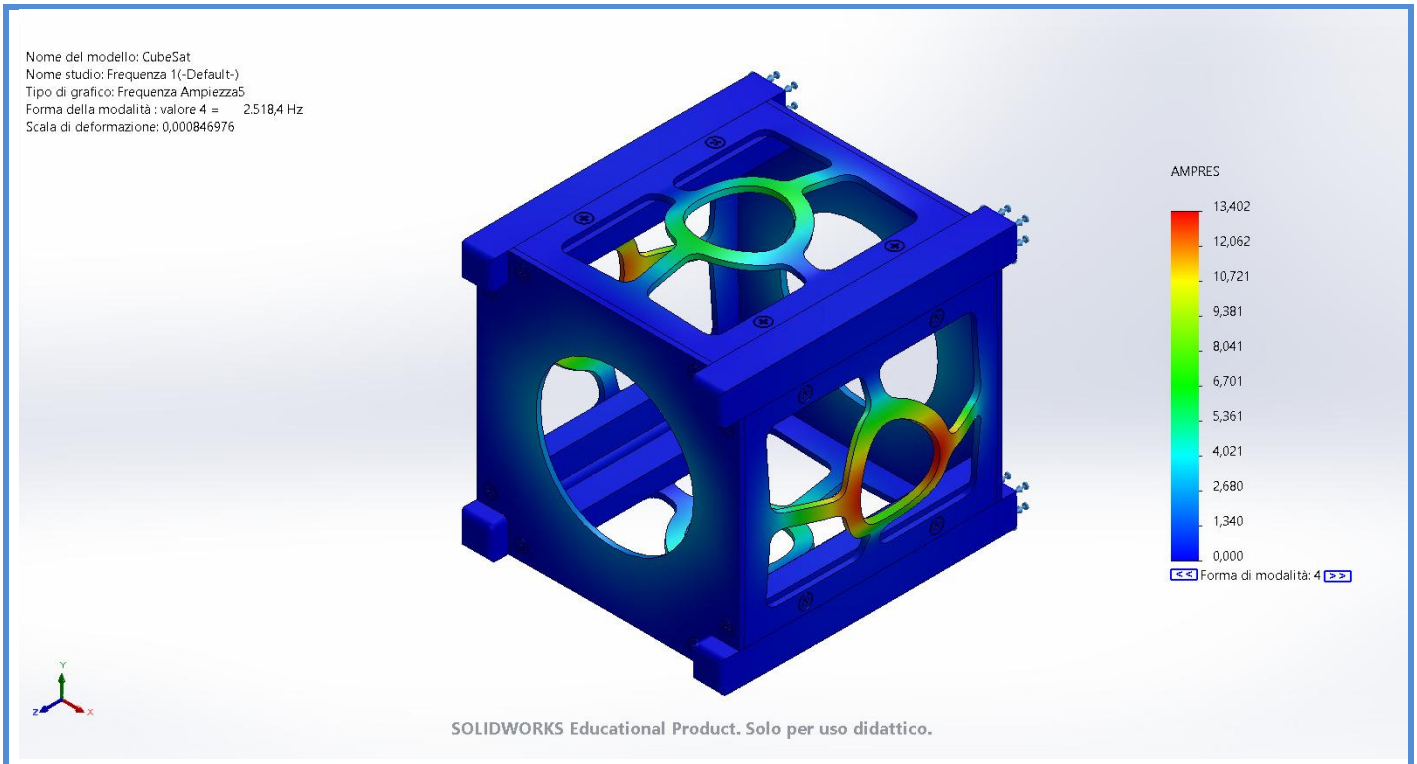
CubeSat-Frequenza 1-Ampiezza-Ampiezza3

Nome	Tipo	Min	Max
Ampiezza4	AMPRES: ampiezza risultante Grafico per Sagoma modale: 4(Valore = 2.518,41 Hz)	0,000 Nodo: 67439	13,402 Nodo: 55889



CubeSat-Frequenza 1-Ampiezza-Ampiezza4

Nome	Tipo	Min	Max
Ampiezza5	AMPRES: ampiezza risultante Grafico per Sagoma modale: 4(Valore = 2.518,41 Hz)	0,000 Nodo: 67439	13,402 Nodo: 55889



CubeSat-Frequenza 1-Ampiezza-Ampiezza5

Elenco modalità

Numero delle frequenze	Rad/sec	Hertz	Secondi
1	6.768,3	1.077,2	0,00092832
2	6.776,2	1.078,5	0,00092724
3	9.449,5	1.503,9	0,00066492
4	15.824	2.518,4	0,00039708
5	15.952	2.538,9	0,00039387

Partecipazione di massa (normalizzata)

N. modo	Frequenza(Hertz)	Direzione X	Direzione Y	Direzione Z
1	1.077,2	0,0044113	0,70326	2,8615e-08
2	1.078,5	0,70338	0,0044109	3,1658e-09
3	1.503,9	1,9363e-06	2,0171e-07	8,98e-09
4	2.518,4	1,4869e-06	1,167e-06	0,0002279

5	2.538,9	8,877e-07	2,0405e-05	3,2342e-05
		Somma X = 0,70779	Somma Y = 0,7077	Somma Z = 0,00026028

Conclusione

L'analisi modale condotta sulla struttura primaria del CubeSat 1U in Alluminio 6061-T6 certifica una prima frequenza naturale pari a 1.077,2 Hz. Tale risultato supera ampiamente la soglia critica di 90-100 Hz imposta dagli standard di lancio aerospaziali. La geometria del telaio si dimostra pertanto estremamente rigida e dinamicamente validata. L'assenza di frequenze proprie a bassi regimi garantisce la totale immunità del satellite ai fenomeni di risonanza distruttiva tipici delle violente vibrazioni generate durante la fase di ascesa del razzo.